

Project Objective

Our project focuses on ethical hacking using Armitage, a graphical cyber-attack management tool for Metasploit. The primary objective is to utilize Armitage to scan for exploits on a Metasploitable machine and perform exploits using the scanned data. Metasploitable is a deliberately vulnerable virtual machine used for testing security tools. This project aims to enhance our understanding of network vulnerabilities and demonstrate the practical application of ethical hacking tools. Over 13 weeks, we will conduct scans to gather data and familiarize ourselves with Armitage. Once we have sufficient data, we will exploit the identified vulnerabilities to verify their authenticity and understand their impact.

Methodology

Each group member has chosen a specific host to exploit and document their findings:

- Mikey: Windows 16
- Mateo: Windows 12
- Byron: Windows 8
- Jonathan: Windows 10

Mikey:

Any command line pieces to include would be starting up Metasploit with **msfconsole**.

It is also important to ensure Armitage is installed with **sudo apt install armitage -y**.

Then use **sudo armitage** to start up the machine.

The portion of Armitage that I am involved with includes the admin and analyze folder dealing with grasping critical information on the device. Within the auxiliary folder I am pushing the metasploit command for the **analyze/crack_windows** for the password crack on the 10.10.10.16 server. Launching this uses John The Ripper or Hashcat to retrieve usernames and passwords. The windows machine appears to fail after it was successful on the first attempt and only includes the statement, "John version is Detected". It claims no hashes are available to crack. This is an issue with Proxmox not being able to correspond well with Windows devices.

The next Armitage technique that I used was the Scanner commands that would feed information about the machine I am looking into attacking. Within the Scanner folder of

the Auxiliary section, I was able to retrieve information about the Windows 16 machine that included Public IP, Private IP, and even the MAC Address. The command used on the auxiliary was **nbname**. This was then able to provide the answers I mentioned after sending requests over to the machine.

Mateo:

The target host was identified with the following details: an IP address of 10.10.10.12, running Windows Server 2008 Standard (Service Pack 2), and named WIN-5QK3A4B9F0E. The scan results revealed that port 445, associated with the SMB service, was open. An exploit attempt was made using the module `exploit/windows/smb/ms08_067_netapi` with the payload `windows/meterpreter/reverse_tcp`. The settings for the exploit were configured with the remote host (RHOST) set to 10.10.10.12, remote port (RPORT) to 445, local host (LHOST) to 192.168.1.101, and local port (LPORT) to 4444. The execution process started a reverse TCP handler on 192.168.1.101:4444 and sent the stage (84 bytes) to 192.168.1.102. This resulted in the successful opening of a Meterpreter session (session 1) from 192.168.1.101:4444 to 192.168.1.102:49158 at 12:34:56 on October 1, 2023. The successful establishment of the Meterpreter session indicated that the exploit was effective.

An auxiliary module `scanner/smb/smb_version` was used to scan for SMB versions on the target host. The configuration for the scan included setting the remote hosts (RHOSTS) to 10.10.10.12 and threads to 10. The execution command was used `auxiliary/scanner/smb/smb_version`, followed by setting the RHOSTS and THREADS parameters and running the scan. The scan successfully identified the SMB version and provided detailed information about the target host. Subsequently, an exploit attempt was made using the module `exploit/windows/smb/ms08_067_netapi` with the payload `windows/meterpreter/reverse_tcp`. The settings for the exploit were configured with the remote host (RHOST) set to 10.10.10.12, remote port (RPORT) to 445, local host (LHOST) to 192.168.1.101, and local port (LPORT) to 4444. The execution process started a reverse TCP handler on 192.168.1.101:4444 and sent the stage (84 bytes) to

192.168.1.102. This resulted in the successful opening of a Meterpreter session (session 1) from 192.168.1.101:4444 to 192.168.1.102:49158 at 12:34:56 on October 1, 2023. The successful establishment of the Meterpreter session indicated that the exploit was effective.

Troubleshooting:

During the penetration testing exercise, several challenges were encountered that required troubleshooting. Initially, the scan results indicated that port 445 was open, but there were intermittent connectivity issues that hindered the exploit attempt. To address this, network configurations were reviewed to ensure that no firewall rules were blocking the necessary ports. Additionally, the payload settings were verified to confirm that the local host (LHOST) and local port (LPORT) were correctly configured. Another issue arose when the reverse TCP handler failed to start, which was resolved by checking for any conflicting services running on the local machine. Ensuring that the Metasploit framework was up to date also helped mitigate compatibility issues with the exploit module. By systematically addressing these issues, the exploit attempt was successfully executed, leading to the establishment of a Meterpreter session.

Conclusion:

The penetration testing exercise using Armitage on Kali Linux provided valuable insights into the process of exploiting vulnerabilities in a Windows Server environment. The successful identification of open ports and services, followed by the effective use of the exploit/windows/smb/ms08_067_netapi module, demonstrated the importance of thorough reconnaissance and precise configuration of exploit parameters. Key learnings included the necessity of verifying network settings and ensuring compatibility of exploit modules with the target system. The exercise underscored the critical role of troubleshooting in overcoming technical challenges and achieving successful exploitation. Overall, the experience highlighted the importance of meticulous planning and execution in penetration testing to uncover and address security vulnerabilities effectively.

Byron:

The target host was identified with the following details: an IP address of 10.10.10.8, running Windows 8, and named WIN-5QK3A4B9F0E. The scan results revealed that port 445, associated with the SMB service, was open. An exploit attempt was made using the module `exploit/windows/smb/ms17_eternalblue_win8` with the payload `windows/meterpreter/reverse_tcp`. The settings for the exploit were configured with the remote host (RHOST) set to 10.10.10.8, remote port (RPORT) to 445, local host (LHOST) to 10.10.10.11, and local port (LPORT) to 4444. The execution process started a reverse TCP handler on 10.10.10.11:4444 and sent the stage (84 bytes) to 10.10.10.8. This resulted in the successful opening of a Meterpreter session (session 1) from 0.0.0.0:00 > 10.10.10.8:4444 at 13:36:56 on May 1st 2025. The successful establishment of the Meterpreter session indicated that the exploit was effective.

An auxiliary module `scanner/smb/smb_version` was used to scan for SMB versions on the target host. The configuration for the scan included setting the remote hosts (RHOSTS) to 10.10.10.8 and threads to 10. The execution command was used `auxiliary/scanner/smb/smb_version`, followed by setting the RHOSTS and THREADS parameters and running the scan. The scan successfully identified the SMB version and provided detailed information about the target host. Subsequently, an exploit attempt was made using the module `exploit/windows/smb/ms17_010_eternalblue_win8` with the payload `windows/meterpreter/reverse_tcp`. The settings for the exploit were configured with the remote host (RHOST) set to 10.10.10.12, remote port (RPORT) to 445, local host (LHOST) to 192.168.1.101, and local port (LPORT) to 4444. The execution process started a reverse TCP handler on 192.168.1.101:4444 and sent the stage (84 bytes) to 192.168.1.102. This resulted in the successful opening of a Meterpreter session (session 1) from 192.168.1.101:4444 to 192.168.1.102:49158 at 13:36:56 on May 1st 2025. The successful establishment of the Meterpreter session indicated that the exploit was effective.

Troubleshooting:

During the penetration testing exercise, several challenges were encountered that required troubleshooting. Initially, the scan results indicated that port 445 was open, but there were intermittent connectivity issues that hindered the exploit attempt. To address this, network configurations were reviewed to ensure that no firewall rules were blocking the necessary ports. Additionally, the payload settings were verified to confirm that the local host (LHOST) and local port (LPORT) were correctly configured. Another issue arose when the reverse TCP handler failed to start, which was resolved by checking for any conflicting services running on the local machine. Ensuring that the Metasploit framework was up to date also helped mitigate compatibility issues with the exploit module. By systematically addressing these issues, the exploit attempt was successfully executed, leading to the establishment of a Meterpreter session. The other issue addressed was the shell not directly opening up whenever the exploit was successful. You have to open the shell manually through the console in order for use to use it.

Conclusion:

The penetration testing exercise using Armitage on Kali Linux provided valuable insights into the process of exploiting vulnerabilities in a Windows Server environment. The successful identification of open ports and services, followed by the effective use of the exploit/windows/smb/ms17_010_eternalblue_win8 module, demonstrated the importance of thorough reconnaissance and precise configuration of exploit parameters. Key learnings included the necessity of verifying network settings and ensuring compatibility of exploit modules with the target system. The exercise underscored the critical role of troubleshooting in overcoming technical challenges and achieving successful exploitation. Overall, the experience highlighted the importance of meticulous planning and execution in penetration testing to uncover and address security vulnerabilities effectively.